United States Patent 12403309
Kind Code B2
Date of Patent September 02, 2025
Inventor(s) Bennett; Robert Graham

Transcutaneous power and data communication link

Abstract

Presented herein are techniques for transcutaneously transferring power and data from an external component to an implantable component of an implantable medical device. In accordance with embodiments presented herein, the implantable component comprises an implantable resonant circuit, while the external component comprises an external resonant circuit. The external component also comprises external radio-frequency (RF) interface circuitry configured to drive the external resonant circuit at a first frequency in order to transfer power to the implantable resonant circuit, and to drive the external resonant circuit at a second frequency, which is different from the first frequency, in order to transfer data to the implantable resonant circuit.

Inventors:	**Bennett; Robert Graham (St Peters, AU)**
Applicant:	**Cochlear Limited** (Macquarie University, AU)
Family ID:	**74210272**
Assignee:	**Cochlear Limited (Macquarie University, AU)**
Appl. No.:	**17/294085**
Filed (or PCT Filed):	**July 10, 2020**
PCT No.:	**PCT/IB2020/056528**
PCT Pub. No.:	**WO2021/009646**
PCT Pub. Date:	January 21, 2021

Prior Publication Data

Document Identifier	**Publication Date**
US 20220072306 A1	Mar. 10, 2022

Related U.S. Application Data

us-provisional-application US 62874117 20190715

Publication Classification

Int. Cl.: **A61N1/36** (20060101); **A61N1/372** (20060101); **A61N1/378** (20060101); **H02J50/12** (20160101); **H02J50/20** (20160101)

U.S. Cl.:

CPC **A61N1/36038** (20170801); **A61N1/37252** (20130101); **A61N1/3787** (20130101); **H02J50/12** (20160201); **H02J50/20** (20160201); H02J2310/23 (20200101)

Field of Classification Search

CPC: F02B (2075/025); F02B (63/00); F02B (75/16); F01P (1/02); A61N (1/36038); A61N (1/37223); A61N (1/37252); A61N (1/3787); F02M (35/024); H02J (2310/23); H02J (50/12); H02J (50/20)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2009/0085408	12/2008	Bruhn	N/A	N/A
2012/0274270	12/2011	Dinsmoor et al.	N/A	N/A
2015/0209591	12/2014	Meskens	N/A	N/A
2016/0285300	12/2015	Summers et al.	N/A	N/A
2017/0127196	12/2016	Blum	N/A	H04R 25/554
2017/0141584	12/2016	DeVaul et al.	N/A	N/A
2017/0246462	12/2016	Meskens	N/A	N/A
2018/0239999	12/2017	Gayton	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
101859484	12/2009	CN	N/A
108781094	12/2017	CN	N/A
H04336801	12/1991	JP	N/A
2017059540	12/2016	WO	N/A

OTHER PUBLICATIONS

International Search Report and Written Opinion in counterpart International Application No. PCT/IB2020/056528, mailed Oct. 13, 2020, 13 pages. cited by applicant
Extended European Search Report in counterpart European Application No. 20840222.2-1202, mailed Aug. 8, 2023, 7 pages. cited by applicant

Primary Examiner: Hulbert; Amanda K

Attorney, Agent or Firm: Edell, Shapiro & Finnan, LLC

Background/Summary

BACKGROUND

Field of the Invention

(1) The present invention relates generally to transcutaneous communication links in implantable medical device systems.

Related Art

(2) Medical device systems having one or more implantable components, generally referred to herein as implantable medical device systems, have provided a wide range of therapeutic benefits to recipients over recent decades. In particular, partially or fully-implantable medical device systems such as hearing prosthesis systems (e.g., systems that include bone conduction devices, mechanical stimulators, cochlear implants, etc.), implantable pacemakers, defibrillators, functional electrical stimulation systems, etc., have been successful in performing lifesaving and/or lifestyle enhancement functions for a number of years.

(3) The types of implantable medical device systems and the ranges of functions performed thereby have increased over the years. For example, many implantable medical devices now often include one or more instruments, apparatus, sensors, processors, controllers or other functional mechanical or electrical components that are permanently or temporarily implanted in a recipient. These functional devices are typically used to diagnose, prevent, monitor, treat, or manage a disease/injury or symptom thereof, or to investigate, replace or modify the anatomy or a physiological process. Many of these functional devices utilize power and/or data received from external devices that are part of, or operate in conjunction with, the implantable medical device system.

SUMMARY

(4) In one aspect an implantable medical device is provided. The implantable medical device comprises: an implantable resonant circuit comprising an implantable coil; an external resonant circuit comprising an external coil configured to transcutaneously transfer power and data to the implantable resonant circuit using separate power and data time slots, respectively; and external radio-frequency (RF) interface circuitry configured to drive the external resonant circuit at a first frequency during the power time slots and to drive the external resonant circuit at a second frequency during data time slots, wherein the second frequency is different from the first frequency.

(5) In another aspect a method is provided. The method comprises: during a first set of time periods, driving an external resonant circuit comprising an external coil with power drive signals having a first center frequency to cause the external coil to transfer power to an implantable resonant circuit; and during a second set of time periods that are different from the first set of time periods, driving the external resonant circuit with data drive signals having a second center frequency to cause the external coil to transfer data to the implantable resonant circuit, wherein the second frequency is different from the first frequency, and wherein the external resonant circuit and the implantable resonant circuit each have an associated tuned frequency that remains the same during each of the first and second sets of time periods.

(6) In another aspect an external component of an implantable medical device is provided. The external component comprises: an external coil configured to forming a transcutaneous communication link with an implantable resonant circuit; power drive circuitry configured to drive the external resonant circuit with power drive signals having a first center frequency to cause the external coil to transfer power to the implantable resonant circuit; and data drive circuitry configured to drive the external resonant circuit with data drive signals having a second center frequency to cause the external coil to transfer power to the implantable resonant circuit, wherein the first frequency provides a selected power coupling between the external resonant circuit and the

implantable resonant circuit, and wherein the second frequency is frequency spaced from the first frequency by a selected frequency distance so as to provide a selected bandwidth for the transcutaneous communication link.
(7) In another aspect a method is provided. The method comprises: sending, via an external resonant circuit of an external component of an implantable medical device, power signals to an implantable resonant circuit of the implantable medical device, wherein the power signals have a first frequency; and sending, via the external resonant circuit, data signals to the implantable resonant circuit, wherein the data signals have a second frequency, and wherein a physical arrangement of each of the implantable resonant circuit and the external resonant circuit remains fixed does not change during either of the power or data time slots.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Embodiments of the present invention are described herein in conjunction with the accompanying drawings, in which:
(2) FIG. **1**A is a schematic diagram illustrating a cochlear implant, in accordance with certain embodiments presented herein;
(3) FIG. **1**B is a block diagram of the cochlear implant of FIG. **1**A;
(4) FIG. **2** is a graph illustrating the relationship between the quality factor and operational frequency of a closely-coupled wireless link, in accordance with certain embodiments presented herein;
(5) FIG. **3** is schematic diagram illustrating a resonant system for use in the transcutaneous transfer of power and data, in accordance with certain embodiments presented herein.
(6) FIG. **4**A is a schematic diagram illustrating portion of an implantable component, in accordance with certain embodiments presented herein;
(7) FIG. **4**B is a schematic diagram illustrating portion of an implantable component, in accordance with certain embodiments presented herein;
(8) FIG. **4**C is a schematic diagram illustrating portions of an external component and an implantable component, in accordance with certain embodiments presented herein;
(9) FIG. **5** is flowchart of a method, in accordance with certain embodiments presented herein;
(10) FIG. **6** is flowchart of another method, in accordance with certain embodiments presented herein; and
(11) FIG. **7** is a schematic diagram illustrating a balance prosthesis in which the techniques presented herein may be implemented.
DETAILED DESCRIPTION
(12) Presented herein are techniques for transcutaneously transferring power and data from an external component to an implantable component of an implantable medical device. In accordance with embodiments presented herein, the implantable component comprises an implantable resonant circuit, while the external component comprises an external resonant circuit. The external component also comprises external radio-frequency (RF) interface circuitry configured to drive the external resonant circuit at a first frequency in order to transfer power to the implantable resonant circuit, and to drive the external resonant circuit at a second frequency, which is different from the first frequency, in order to transfer data to the implantable resonant circuit.
(13) There are a number of different types of implantable medical device systems in which embodiments presented herein may be implemented. However, merely for ease of illustration, the techniques presented herein are primarily described with reference to one type of implantable medical device system, namely a cochlear implant. It is to be appreciated that the techniques presented herein may be used in any other partially or fully implantable medical devices now

known or later developed, including other auditory prostheses, such as auditory brainstem stimulators, electro-acoustic hearing prostheses, acoustic hearing aids, bone conduction devices, middle ear prostheses, direct cochlear stimulators, bimodal hearing prostheses, etc. The techniques presented herein may also be used with balance prostheses (e.g., vestibular implants), retinal or other visual prosthesis/stimulators, occipital cortex implants, sensor systems, implantable pacemakers, drug delivery systems, defibrillators, catheters, seizure devices (e.g., devices for monitoring and/or treating epileptic events), sleep apnea devices, electroporation devices, spinal cord stimulators, deep brain stimulators, motor cortex stimulators, sacral nerve stimulators, pudendal nerve stimulators, vagus/vagal nerve stimulators, trigeminal nerve stimulators, diaphragm (phrenic) pacers, pain relief stimulators, other neural, neuromuscular, or functional stimulators, etc.
(14) FIG. **1**A is a schematic diagram of an exemplary cochlear implant **100** in accordance with aspects presented herein, while FIG. **1**B is a block diagram of the cochlear implant **100**. For ease of illustration, FIGS. **1**A and **1**B will be described together.
(15) The cochlear implant **100** comprises an external component **102** and an internal/implantable component **104**. The external component **102** is directly or indirectly attached to the body of the recipient and typically comprises an external coil **106** and, generally, a magnet (not shown in FIG. **1**) fixed relative to the external coil **106**. The external component **102** also comprises one or more input elements/devices **113** for receiving input signals at a sound processing unit **112**. In this example, the one or more input devices **113** include sound input devices **108** (e.g., microphones positioned by auricle **110** of the recipient, telecoils, etc.) configured to capture/receive input signals, one or more auxiliary input devices **109** (e.g., audio ports, such as a Direct Audio Input (DAI), data ports, such as a Universal Serial Bus (USB) port, cable port, etc.), and a wireless transmitter/receiver (transceiver) **111**, each located in, on, or near the sound processing unit **112**.
(16) The sound processing unit **112** also includes, for example, at least one battery **107**, external radio-frequency (RF) interface circuitry **121**, and a processing module **125**. The processing module **125** may comprise a number of elements, including a sound processor **131**. As described further below, the external RF interface circuitry **121** comprises data drive circuitry **144** and power drive circuitry **146** which are selectively activated/used for transcutaneous transmissions of data and power, respectively, to the implantable component **104**.
(17) In the examples of FIGS. **1**A and **1**B, the sound processing unit **112** is a behind-the-ear (BTE) sound processing unit configured to be attached to, and worn adjacent to, the recipient's ear. However, it is to be appreciated that embodiments of the present invention may be implemented by sound processing units having other arrangements, such as by an off-the-ear (OTE) sound processing unit (i.e., a component having a generally cylindrical shape and which is configured to be magnetically coupled to the recipient's head), etc., a mini or micro-BTE unit, an in-the-canal unit that is configured to be located in the recipient's ear canal, a body-worn sound processing unit, etc.
(18) Returning to the example embodiment of FIGS. **1**A and **1**B, the implantable component **104** comprises an implant body (main module) **114**, a lead region **116**, and an intra-cochlear stimulating assembly **118**, all configured to be implanted under the skin/tissue (tissue) **105** of the recipient. The implant body **114** generally comprises a hermetically-sealed housing **115** in which internal RF interface circuitry **124**, a power supply **129** (e.g., one or more implantable batteries, one or more capacitors, etc.), and a stimulator unit **120** are disposed. The stimulator unit **120** comprises, among other elements, one or more current sources on an integrated circuit (IC).
(19) The implant body **114** also includes an internal/implantable coil **122** that is generally external to the housing **115**, but which is connected to the RF interface circuitry **124** via a hermetic feedthrough (not shown in FIG. **1**B). It is to be appreciated that implantable component **104** and/or the external component **102** may include other components that, for ease of illustration, have been omitted from FIGS. **1**A and **1**B.
(20) As noted, the cochlear implant **100** includes the external coil **106** and the implantable coil **122**.

The coils **106** and **122** are typically wire antenna coils each comprised of multiple turns of electrically insulated single-strand or multi-strand platinum or gold wire. Generally, a magnet is fixed relative to each of the external coil **106** and the implantable coil **122**. The magnets fixed relative to the external coil **106** and the implantable coil **122** facilitate the operational alignment of the external coil with the implantable coil.

(21) The operational alignment of the coils **106** and **122** enables the external component **102** to transfer power (e.g., for use in powering components of the implantable component) and data (e.g., for use in generating signal signals) to the implantable component **104** via a bidirectional "transcutaneous communication link" or "closely-coupled wireless link" **127** formed between the external coil **106** with the implantable coil **122**. That is, due to the operational alignment, the data drive circuitry **144** in external RF interface circuitry **121** can be used to transfer data to the implantable component **104** via the closely-coupled wireless link **127**. Similarly, the operational alignment of coils **106** and **122** enables the power drive circuitry **146** to transfer power signals (power) to the implantable component **104** via the closely-coupled wireless link **127**. The power signals, when received by the internal RF interface circuitry **124**, may be used to power the elements of implantable component **104** and/or used to provide power to the power supply **129**.

(22) In certain examples, the closely-coupled wireless link is a radio frequency (RF) link. However, various other types of energy transfer, such as infrared (IR), electromagnetic, capacitive and inductive transfer, may be used to transfer the power and/or data from an external component to an implantable component and, as such, FIG. **1**B illustrates only one example arrangement.

(23) As noted above, sound processing unit **112** includes the processing module **125**. The processing module **125** is configured to convert input audio signals into stimulation control data **136** for use in stimulating a first ear of a recipient (i.e., the processing module **125** is configured to perform sound processing on input audio signals received at the sound processing unit **112**). Stated differently, the sound processor **131** (e.g., one or more processing elements implementing firmware, software, etc.) is configured to convert the captured input audio signals into stimulation control data **136** that represents stimulation signals for delivery to the recipient. The input audio signals that are processed and converted into stimulation control data may be audio signals received via the sound input devices **108**, signals received via the auxiliary input devices **109**, and/or signals received via the wireless transceiver **111**.

(24) In the embodiment of FIG. **1**B, the stimulation control data **136** is provided to the external RF interface circuitry **121**, where the data drive circuitry **144** transcutaneously transfers the stimulation control data **136** (e.g., in an encoded manner) to the implantable component **104** via external coil **106** and implantable coil **122**. That is, the stimulation control data **136** is sent by the data drive circuitry **144** over the closely-coupled wireless link **127**. The internal RF interface circuitry **124** is configured to receive the stimulation control data **136** via implantable coil **122** and to provide that data to the stimulator unit **120**. The stimulator unit **120** is configured to utilize the stimulation control data **136** to generate stimulation signals (e.g., current signals) for delivery to the recipient's cochlea via the stimulating assembly **118**. In this way, cochlear implant **100** electrically stimulates the recipient's auditory nerve cells, bypassing absent or defective hair cells that normally transduce acoustic vibrations into neural activity, in a manner that causes the recipient to perceive one or more components of the input audio signals.

(25) More specifically, as noted above, stimulating assembly **118** is configured to be at least partially implanted in the recipient's cochlea **140**. Stimulating assembly **118** includes a plurality of longitudinally spaced intra-cochlear electrical contacts (electrode contacts or electrodes) **126** that collectively form an electrode contact array **128** configured to, for example, deliver electrical stimulation signals (current signals) generated based on the stimulation control data **136** to the recipient's cochlea. In certain examples, the electrode contacts **126** may also be used to sink stimulation signals from the recipient's cochlea.

(26) FIG. **1**A illustrates a specific arrangement in which stimulating assembly **118** comprises

twenty-two (22) intra-cochlear electrode contacts **126**, labeled as electrode contacts **126**(**1**) through **126**(**22**). It is to be appreciated that embodiments presented herein may be implemented in alternative arrangements having different numbers of intra-cochlear electrode contacts.
(27) As shown, the intra-cochlear electrode contacts **126**(**1**)-**126**(**22**) are disposed in an elongate carrier member **134**. The carrier member **134** has a center longitudinal axis and an outer surface. The carrier member **134** is formed from a non-conductive (insulating) material, such as silicone or other elastomer polymer. As such, the carrier member **134** electrically isolates the intra-cochlear electrode contacts **126**(**1**)-**126**(**22**) from one another. As shown in FIG. **1**B, the intra-cochlear electrode contacts **126**(**1**)-**126**(**22**) are each spaced from one another by sections/segments of the carrier member **134**.
(28) The stimulating assembly **118** extends through an opening in the recipient's cochlea (e.g., cochleostomy, the round window, etc.) and has a proximal end connected to stimulator unit **120** via lead region **116** and a hermetic feedthrough (not shown in FIG. **1**B). Carrier member **134** and lead region **116** each includes a plurality of conductors (wires) extending there through that electrically connect the electrode contacts **126** to the stimulator unit **120**.
(29) Also shown in FIG. **1**A is an extra-cochlear electrode contact **126**(**23**). The extra-cochlear electrode contact **126**(**23**) is an electrical contact that is configured to, for example, deliver electrical stimulation to the recipient's cochlea and/or to sink current from the recipient's cochlea. The extra-cochlear electrode contact **126**(**23**) is connected to a reference lead **123** that includes one or more conductors that electrically couple the extra-cochlear electrode contact **126**(**23**) to the stimulator unit **120**.
(30) As noted above, the closely-coupled wireless link **127** formed between the external coil **106** and the implantable coil **122** may be used to transfer power and/or data from the external component **102** to the implantable component **104**. In certain examples, the power and data are transmitted using a type of time division multiple access (TDMA) technique to share the closely-coupled wireless link **127**. That is, the closely-coupled wireless link **127** is used to separately transfer power and data from the external component **102** to the implantable component **104**, where the transfer of power and data occur during separate (different and non-overlapping) time slots using the same external coil **106** (i.e., a shared external coil for both data and power). For example, during a set of first time periods, the power drive circuitry **146** of the external RF interface circuitry **121** is configured to drive (energize) the external coil **106** in a manner that sends data to the implantable component **104**. During a second set of time periods, the data drive circuitry **144** of the external RF interface circuitry **121** is configured to drive (energize) the external coil **106** in a manner that sends power to the implantable component **104**. A single transmission sequence/frame may be split into a power time slot (block) and a data time slot (block) and repeated. All of the power towards the implantable component **104** is transferred during the power time slot.
(31) In the example of FIGS. **1**A and **1**B, the external coil **106** is part of an external resonant circuit (e.g., external resonant tank circuit) **140**. Similarly, the implantable coil **122** and at least a portion of the internal RF interface circuitry **124** form an implantable resonant circuit (e.g., internal resonant tank circuit) **142**. The external resonant tank circuit **140** and the internal resonant tank circuit **142** collectively form a resonant system **150** which function as the bidirectional closely-coupled wireless link **127**.
(32) One measure of the operation of the closely-coupled wireless link **127** (i.e., of the resonant system **150** formed by the external resonant tank circuit **140** and the internal resonant tank circuit **142**) is the quality factor (Q) of the link. In general, the quality factor is a ratio of power stored to power dissipated in the resonant system reactance and resistance, respectively. The quality factor is a dimensionless number that describes the damping in the resonant system, as well as provides an indication of the bandwidth relative to the center frequency. A higher value corresponds to a narrower bandwidth.
(33) Returning to FIGS. **1**A and **1**B, in order to efficiently transfer power from the external

component to **102** to the implantable component **104**, the closely-coupled wireless link **127** (resonant system **150**) should have a high quality factor. That is, the quality factor of the closely-coupled wireless link **127** should be maximized during power transmission, thereby ensuring low power loss. However, as noted, a high quality factor is associated with a narrow bandwidth, which is problematic for transmission of data over the closely-coupled wireless link **127**. Therefore, power transmission and data transmission have competing quality factor requirements (i.e., efficient power transmission requires a high/maximum quality factor, while higher bandwidth data requires a lower quality factor).
(34) The techniques presented herein address these competing quality factor requirements for power and data transmissions through the use of different transmit (drive) frequencies at the external resonant circuit **140**. More specifically, in the embodiments of FIGS. **1**A and **1**B, the power drive circuitry **146** is configured to drive the external coil **106** (external resonant inductive coil) at a first frequency to transmit power over the closely-coupled wireless link **127**. Both the external resonant circuit **140** and the implantable resonant circuit **142** are substantially tuned to this same first frequency. That is, the external resonant circuit **140** and the implantable resonant circuit **142** are each structurally configured so as to resonate a frequency that is substantially the same as the first frequency. Accordingly, the resonant system **150** may be referred to as being tuned to the first frequency. In other words, in these embodiments, the first frequency for power transmission is the resonant frequency of the resonant system **150** (i.e., the resonant frequency of each of the external resonant circuit **140** and the implantable resonant circuit **142**).
(35) Due to the fact that the power transmissions occur at a frequency that substantially matches the tuned frequency of each of the external resonant circuit **140** and the implantable resonant circuit **142** (i.e., the tuned frequency of the resonant system **150**), maximum power coupling is achieved with the power transmissions at the first frequency. Stated differently, the matching of the drive/transmit frequency to the tuned frequency of the resonant system **150** provides a high quality factor where, as noted, the higher the quality factor of the system, the more efficient the power transfer will be across the closely-coupled wireless link **127**.
(36) While, as noted above, a high quality factor is appropriate for power transmission, a high quality factor reduces the rate that data can be transmitted through the inductive coupling of the closely-coupled wireless link **127** (i.e., reduces the available bandwidth of the closely-coupled wireless link **127**). While the quality factor of the resonant system is high when the transmit frequency is at or near the resonant frequency, the quality factor is lower at different frequencies that have an appropriate distance/spacing, in frequency, from the resonant frequency (where the frequency difference is dependent on the shape of the resonance and is selected to provide an appropriate bandwidth for the desired data rate).
(37) Accordingly, an appropriate quality factor for transmitting data can be obtained at transmit frequencies that are spaced some frequency distance from the resonant frequency of the resonant system **150**. Therefore, in accordance with embodiments presented herein, data drive circuitry **144** is configured to drive the external resonant circuit **140**, including external coil **106**, at a second frequency to transmit data over the closely-coupled wireless link **127**, where the second frequency is different from the first frequency. During the data transmission, the external resonant circuit **140** and the implantable resonant circuit **142** both remain tuned to the first frequency (i.e., the external resonant circuit **140** and the implantable resonant circuit **142** each have a fixed structure that fixes the tuned frequency thereof). As such, the frequency “mismatch” or difference between the transmit frequency and the frequency of the resonant system **150** causes a reduction in the quality factor of the combined resonant system (i.e., reduces the quality factor of the closely-coupled wireless link **127**), which in turn increases the bandwidth available for the transmission of the data.
(38) In summary, FIGS. **1**A and **1**B illustrate an arrangement in accordance with embodiments presented herein in which, during a first set of time periods, the external resonant circuit **140**, which includes external coil **106**, is driven at a first frequency to transmit power to the implantable

resonant circuit **142**, including implantable coil **122**. During a second set of time periods, the external resonant circuit **140** is driven at a second frequency to transmit data to the implantable resonant circuit **142**, where the second frequency is frequency spaced a frequency distance from the first frequency. During both the first and second sets of time periods, the external resonant circuit **140** and the implantable resonant circuit **142** remain tuned to the first frequency (i.e., the external resonant circuit **140** and the implantable resonant circuit **142** have a fixed tuning).

(39) FIG. **2** is a graph **250** illustrating the relationship between the quality factor and frequency of a bidirectional closely-coupled wireless link, such as closely-coupled wireless link **127**. In particular, graph **250** includes a vertical (Y) axis **252** illustrating the quality factor of a closely-coupled wireless link, and a horizontal (X) axis **254** representing the transmit frequency of the closely-coupled wireless link. As represented by line **256**, the quality factor is maximized (i.e., is the highest) when the transmit frequency (f) (e.g., the frequency at which signals are transmitted by an external coil) is substantially the same as the resonant frequency (f'') of the closely-coupled wireless link. As shown by line **258**, the quality factor is reduced when the transmit frequency (f) is lower than the resonant frequency (f'') of the closely-coupled wireless link (e.g., the Q is lower when $f=f''/2$). Similarly, as shown by line **260**, the quality factor is also reduced when the transmit frequency (f) is higher than the resonant frequency (f'') of the closely-coupled wireless link (e.g., the Q is lower when $f=f''*2$).

(40) FIG. **3** is schematic diagram illustrating a resonant system **350** for use in the transcutaneous transfer of power and data, in accordance with embodiments presented herein. As shown, the resonant system **350** includes an external resonant circuit **340** comprising, among other elements, an external coil **306**. The resonant system **350** also includes an implantable resonant circuit **342** comprising, among other elements, an internal coil **322**. Electrically coupled to, and potentially forming part of, the implantable resonant circuit **342** is internal RF interface circuitry **324**, only a portion of which is shown in FIG. **3**. The resonant system **350** functions as a closely-coupled wireless link, generally illustrated by arrow **327**.

(41) Electrically coupled to the external resonant circuit **340** is external RF interface circuitry **321**, only a portion of which is shown in FIG. **3**. The external RF interface circuitry **321** comprises, among other elements, data drive circuitry **344**, power drive circuitry **346**, and a controller **348**. The data drive circuitry **344** and power drive circuitry **346** may be selectively activated/used, for example under the control of controller **348**, for transcutaneous data and power transmissions via external resonant circuit **340**.

(42) More specifically as noted above, in certain examples power and data are transmitted using a type of time division multiple access (TDMA) technique to share the bidirectional closely-coupled wireless link **327** formed by resonant system **350** (i.e., the closely-coupled wireless link **327** is used to separately transfer power and data from the external component **102** to the implantable component **104**, where the transfer of power and data occur during separate time slots using the same external coil **306**). Therefore, during a set of first time periods, the power drive circuitry **346** is configured to drive (energize) the external coil **306** with power drive signals **364**. The power drive signals **364** comprise an alternating waveform having a steady base frequency of alternation (i.e., a constant burst of square wave at the frequency of resonance of the coil). The frequency of alternation of the power drive signals **364** is sometimes referred to herein as the “power transmission frequency” or the “first frequency.” The first frequency of the power drive signals **364** corresponds to a resonant frequency of the resonant system **350**. That is, the first frequency may be substantially the same as the resonant frequency of the resonant system **350**.

(43) When the coil **306** is driven with the power drive signals **364**, current flow is induced in the implantable coil **322**, where the current flow corresponds to (i.e., represents) the power drive signals **364**. As such, via the inductive link between coils **306** and **322**, the power drive signals **364** are received at the internal RF interface circuitry **324**. The internal RF interface circuitry **324** is configured to direct the power drive signals **364** to, for example, an implantable rechargeable

battery and/or other components. For ease of illustration, the various components configured to receive the power drive signals **364** are collectively and generally represented in FIG. **3** by load **363**.

(44) During a second set of time periods, the data drive circuitry **344** is configured to drive (energize) the external coil **306** with data drive signals **362** in a manner that sends data to the implantable component. The data drive signals **362** comprise the data to be transmitted (e.g., stimulation control data) that is encoded (modulated) onto a carrier signal (i.e., an alternating waveform having a steady base frequency of alternation), where the carrier signal has a second frequency. The frequency of the data carrier signals (i.e., the frequency of the data drive signals **362**) is frequency spaced from the resonant frequency of the resonant system **350**, and is sometimes referred to herein as the "data transmission frequency" or the "second frequency."

(45) The data transmission frequency of the data drive signals **362** is frequency spaced a sufficient distance from the resonant frequency of the link **327** to provide the appropriate quality factor for high bandwidth frequency. The data transmission frequency can be higher or lower than the resonant frequency. In certain embodiments, the data transmission frequency may be a multiple or a division of the resonant frequency.

(46) When the coil **306** is driven with the data drive signals **362**, current flow is induced in the implantable coil **322**, where the current flow corresponds to (i.e., represents) the data drive signals **362**. As such, via the inductive link between coils **306** and **322**, the data drive signals **362** are received at the internal RF interface circuitry **324**. The internal RF interface circuitry **324** is configured to direct the data drive signals **362** to a data output **365** to which any of a number of other components, which have been omitted from FIG. **3** for ease of illustration, may be connected.

(47) As shown, the data drive circuitry **344** and the power drive circuitry **346** are connected to the external resonant circuit **340** via a driver circuit **368**, of which a number of different arrangements is possible. In the example of FIG. **3**, the driver circuit **368** comprises a switch **372** and an amplifier **374**. However, it is to be appreciated that the arrangement for driver circuit **368** shown in FIG. **3** is merely illustrative and that a driver circuit in accordance with embodiments presented herein may have any of a number of different arrangements.

(48) In FIG. **3**, the data drive signals **362** and the power drive signals **364** comprise two inputs to the driver circuit **368**. The switch **372** operates under the control of controller **348** (i.e., control signal **370**) to selectively enable the data drive signals **362** or the power drive signals **364** to pass to the amplifier **374**.

(49) In summary, FIG. **3** illustrates an arrangement in which the external resonant circuit **340** is tuned to a frequency used to transmit power signals, and in which the implantable resonant circuit **342** is tuned to the same frequency for maximum power coupling. In this example, the tuned frequency of the external resonant circuit **340** and the tuned frequency of the implantable resonant circuit **342** are each fixed during transmission of both the power (i.e., when driving the coil **306** with the power drive signals **364**) and the data (i.e., when driving the coil **306** with the data drive signals **362**) (i.e., a fixed resonance for both transmitter and receiver of the link). Accordingly, in accordance with the techniques presented herein, there is no switching of components into or out of either the external resonant circuit **340** or the implantable resonant circuit **342** to change the tuned frequencies or Q factors of the circuits, thereby reducing complexity of the internal and/or external resonant circuitry

(50) Although the tuned frequencies of the external resonant circuit **340** and the implantable resonant circuit **342** are fixed, the frequency of transmission of the power and data signals is switched, where the power phase is transmitted at the resonant frequency of the link. The data phase is transmitted at a frequency different from the resonant frequency, far enough from the resonant frequency to provide the appropriate Q for high bandwidth frequency. The data frequency can be higher or lower than the resonant frequency, and the resonant frequency can be any frequency, but may be chosen as the as one of the ISM (Industrial, Scientific and Medical)

frequency bands where higher electromagnetic (EM) emissions are allowed.
(51) In the example of FIG. **3**, the external resonant circuit **340** and the implantable resonant circuit **342** are "pre-tuned" to the fixed power transmission frequency by design, during manufacture, etc. It is to be noted that the tuning of a coupled system of inductive coils is different from when the coils when uncoupled. Therefore, as used herein, reference to the first frequency (power transmission frequency) or resonant frequency is the frequency that achieves the best power transfer when the external resonant circuit **340** and the implantable resonant circuit **342** are coupled with one another.
(52) While in the embodiment of FIG. **3** the frequencies of the external resonant circuit **340** and the implantable resonant circuit **342** are fixed and pre-tuned (e.g., by design, during manufacture, etc.), FIG. **4**A is a schematic diagram illustrating an alternative embodiment in which at least one of an external resonant circuit or an implantable resonant circuit is self-tuning to the power transmission frequency.
(53) More specifically, shown in FIG. **4**A is a portion of an implantable component **404**(A), including an implantable resonant circuit **442**(A) and internal RF interface circuitry **424**(A). The implantable resonant circuit **442**(A) includes, among other elements, an implantable coil **422**(A). The implantable resonant circuit **442**(A) is configured to form a resonant system with an external resonant circuit (not shown in FIG. **4**A). The resonant system provides a closely-coupled wireless link **427**(A) over which power and data may be sent by an external component (also not shown in FIG. **4**A) to the implantable component **404**(A). The external resonant circuit may have an arrangement that is similar to the arrangement shown in FIG. **3**.
(54) As noted above, resonant systems in accordance with embodiments presented herein that provide a closely-coupled wireless link, such as link **427**(A), are designed to be tuned to maximum power coupling. That is, in accordance with embodiments presented herein, during operation, each of the external resonant circuit and the implantable resonant circuit **442**(A) are configured to be tuned to substantially the same first frequency, where the first frequency provides a high quality factor. Power signals are then transmitted over the closely-coupled wireless link **427**(A) at this same first frequency.
(55) Whereas in FIG. **3** the external resonant circuit and the implantable resonant circuit are pre-tuned to the substantially same first frequency, in the arrangement of FIG. **4**A the implantable component **404**(A) is configured to dynamically tune the implantable resonant circuit **442**(A) to the first frequency. That is, in the example of FIG. **4**A, the external resonant circuit has a pre-tuned frequency. Once the implantable resonant circuit **442**(A) is coupled to the external resonant circuit (i.e., so as to form the resonant system), the implantable component **404**(A) can determine an appropriate tuning for the implantable resonant circuit **442**(A).
(56) In the example of FIG. **4**A, the implantable component **404**(A) includes a control circuit **480**(A) that is able to adjust the frequency of resonance (i.e., the tuned frequency) of the implantable resonant circuit **442**(A). For example, in the arrangement of FIG. **4**A, the implantable resonant circuit **442**(A) includes, among other elements, one or more variable capacitance components **481**(A) collectively having a capacitance that is controlled/set by the control circuit **480**(A). The capacitance of the one or more variable capacitance components **481**(A) can be adjusted, for example, in an analog manner, with digital chips designed to switch different capacitors into the circuit, or in another manner. By adjusting the capacitance of the one or more variable capacitance components **481**(A) the control circuit **480**(A) can adjust the resonant (tuned) frequency of the implantable resonant circuit **442**(A) either up or down. Using the power and/or data signals **482**(A) received at the implantable resonant circuit **442**(A), the control circuit **480**(A) can determine the point of maximum power coupling, and accordingly the resonant frequency of the implantable resonant circuit **442**(A) at that time. The control circuit **480**(A) can then set (i.e., fix) the implantable resonant circuit **442**(A) to the correct tuned frequency (i.e., fix the capacitance of the one or more variable capacitance components **481**(A) to a level that achieves the selected

tuned frequency for implantable resonant circuit **442**(A)). These examples may be advantageous in that, during manufacturing of the system, there will be no "set tuning" stage for at least the implantable resonant circuit **442**(A).

(57) As noted above, in the example of FIG. **4**A, the control circuit **480**(A) is configured determine the tuned frequency of the implantable resonant circuit **442**(A) that provides a maximum power coupling with the external resonant circuit (i.e., when the implantable resonant circuit **442**(A) is tuned to a frequency that substantially matches the tuned frequency of the external resonant circuit). In certain examples, the control circuit **480**(A) includes or is coupled to a measurement circuit **483**(A) that can be used to measure the voltage of the received signals **482**(A) or to use the received signals **482**(A) to determine the power being drawn over the RF link (e.g., switching a resistor and measuring the rectified voltage enables determination of the amount of power drawn). In certain embodiments, the control circuit **480**(A) may adjust the capacitance of the one or more variable capacitance components **481**(A) to increase or decrease the tuned frequency of the implantable resonant circuit **442**(A) in a manner that increases the power measured at the measurement circuit **483**(A). The control circuit **480**(A) continues this adjustment until a decrease in the power measured at the measurement circuit **483**(A) is detected, at which point the control circuit **480**(A) reverses the adjustment to again increase the power. Using increasingly smaller up and down adjustments to the tuned frequency (i.e., to the capacitance of the one or more variable capacitance components **481**(A)), the control circuit **480**(A) can accurately lock to the correct tuned frequency (e.g., to a substantially same frequency as that of the external resonant circuit). Once this dynamic tuning is completed, the implantable resonant circuit **442** remains tuned to the tuned frequency (e.g., the same first) during receipt of both power and data from the external component.

(58) FIG. **4**A illustrates an example in which the control circuit **480**(A) determines the tuned frequency for the implantable resonant circuit **442**(A) by determining the point of maximum power coupling. In an alternative embodiment, the control circuit **480**(A) could directly measure the frequency of the received signals **482**(A) (i.e., identify the pre-tuned frequency of the external resonant circuit) and then adjust the capacitance of the one or more variable capacitance components **481**(A) of the implantable resonant circuit **442** based thereon (e.g., so that the implantable resonant circuit **442** is tuned to the substantially same first frequency).

(59) As noted, in the example of FIG. **4**A, the control circuit **480**(A) is configured to determine the point of maximum power coupling for the resonant system formed by the implantable resonant circuit **442**(A) and an external resonant circuit, and accordingly the resonant frequency of the implantable resonant circuit **442**(A) at that time. In an alternative arrangement, the control circuit **480**(A) could adjust the frequency, not to maximum power coupling, but to a non-optimized power coupling. As used herein, a non-optimized power coupling is power coupling that is lower the maximum power coupling, but which is suitable to power the implantable component and/or is suitable to protect against the receipt of too much power. For example, the use of a non-optimized power coupling could prevent damage to the internal component **404**(A) when an external component sends and there is no feedback to the external component (i.e., de-tuning the implantable resonant circuit **442**(A) results in greater power loss across the link, meaning there is less power received).

(60) As noted, in the example of FIG. **4**A, the control circuit **480**(A) adjusts the capacitance of the one or more variable capacitance components **481**(A) to adjust/change the resonant (tuned) frequency of the implantable resonant circuit **442**(A). It is to be appreciated that this technique for adjusting the resonant frequency of the internal resonant is merely illustrative and that the resonant frequency of an implantable resonant circuit may be changed in other manners.

(61) For example, FIG. **4**B is a schematic diagram of a portion of an implantable component **404**(B), including an implantable resonant circuit **442**(B) and internal RF interface circuitry **424**(B). The implantable resonant circuit **442**(B) includes, among other elements, an implantable coil **422**(A). The implantable resonant circuit **442**(B) is configured to form a resonant system with

an external resonant circuit (not shown in FIG. **4**B). The resonant system provides a closely-coupled wireless link **427**(B) over which power and data may be sent by an external component (also not shown in FIG. **4**B) to the implantable component **404**(B). The external resonant circuit may have an arrangement that is similar to the arrangement shown in FIG. **3**.

(62) As noted above, resonant systems in accordance with embodiments presented herein that provide a closely-coupled wireless link, such as link **427**(B), are designed to be tuned to maximum power coupling. That is, in accordance with embodiments presented herein, during operation, each of the external resonant circuit and the implantable resonant circuit **442**(B) are configured to be tuned to substantially the same first frequency, where the first frequency provides a high quality factor. Power signals are then transmitted over the closely-coupled wireless link **427**(B) at this same first frequency.

(63) Whereas in FIG. **3** the external resonant circuit and the implantable resonant circuit are pre-tuned to the substantially same first frequency, in the arrangement of FIG. **4**B the implantable component **404**(B) is configured to dynamically tune the implantable resonant circuit **442**(B) to the first frequency. That is, in the example of FIG. **4**B, the external resonant circuit has a pre-tuned frequency. Once the implantable resonant circuit **442**(B) is coupled to the external resonant circuit (i.e., so as to form the resonant system), the implantable component **404**(B) can determine an appropriate tuning for the implantable resonant circuit **442**(B).

(64) In the example of FIG. **4**B, the implantable component **404**(B) includes a control circuit **480**(B) that is able to adjust the frequency of resonance (i.e., the tuned frequency) of the implantable resonant circuit **442**(B). For example, in the arrangement of FIG. **4**B, the implantable resonant circuit **442**(B) includes, among other elements, one or more variable inductance components **485**(B) collectively having an inductance that is controlled/set by the control circuit **480**(B). The inductance of the one or more variable inductance components **485**(B) can be adjusted, for example, in an analog manner, with digital chips designed to switch different inductors into the circuit, or in another manner. By adjusting the inductance of the one or more variable inductance components **485**(B) the control circuit **480**(B) can adjust the resonant (tuned) frequency of the implantable resonant circuit **442**(B) either up or down. Using the power and/or data signals **482**(B) received at the implantable resonant circuit **442**(B), the control circuit **480**(B) can determine the point of maximum power coupling, and accordingly the resonant frequency of the implantable resonant circuit **442**(B) at that time. The control circuit **480**(B) can then set (i.e., fix) the implantable resonant circuit **442**(B) to the correct tuned frequency (i.e., fix the capacitance of the one or more variable inductance components **485**(B) to a level that achieves the selected tuned frequency for implantable resonant circuit **442**(B)). These examples may be advantageous in that, during manufacturing of the system, there will be no "set tuning" stage for at least the implantable resonant circuit **442**(B).

(65) As noted above, in the example of FIG. **4**B, the control circuit **480**(B) is configured determine the tuned frequency of the implantable resonant circuit **442**(B) that provides a maximum power coupling with the external resonant circuit (i.e., when the implantable resonant circuit **442**(B) is tuned to a frequency that substantially matches the tuned frequency of the external resonant circuit). In certain examples, the control circuit **480**(B) includes or is coupled to a measurement circuit **483**(B) that can be used to measure the voltage of the received signals **482**(B) or to use the received signals **482**(B) to determine the power being drawn over the RF link (e.g., switching a resistor and measuring the rectified voltage enables determination of the amount of power drawn). In certain embodiments, the control circuit **480**(B) may adjust the inductance of the one or more variable inductance components **485**(B) to increase or decrease the tuned frequency of the implantable resonant circuit **442**(B) in a manner that increases the power measured at the measurement circuit **483**(B). The control circuit **480**(B) continues this adjustment until a decrease in the power measured at the measurement circuit **483**(B) is detected, at which point the control circuit **480**(B) reverses the adjustment to again increase the power. Using increasingly smaller up

and down adjustments to the tuned frequency (i.e., to the inductance of the one or more variable inductance components **485**(B)), the control circuit **480**(B) can accurately lock to the correct tuned frequency (e.g., to a substantially same frequency as that of the external resonant circuit). Once this dynamic tuning is completed, the implantable resonant circuit **442** remains tuned to the tuned frequency (e.g., the same first) during receipt of both power and data from the external component.

(66) FIG. **4**A illustrates an example in which the control circuit **480**(B) determines the tuned frequency for the implantable resonant circuit **442**(B) by determining the point of maximum power coupling. In an alternative embodiment, the control circuit **480**(B) could directly measure the frequency of the received signals **482**(B) (i.e., identify the pre-tuned frequency of the external resonant circuit) and then adjust the capacitance of the one or more variable inductance components **485**(B) of the implantable resonant circuit **442** based thereon (e.g., so that the implantable resonant circuit **442** is tuned to the substantially same first frequency).

(67) As noted, in the example of FIG. **4**A, the control circuit **480**(A) adjusts the capacitance of the one or more variable capacitance components **481**(A) to adjust/change the resonant (tuned) frequency of the implantable resonant circuit **442**(A). In FIG. **4**B, the control circuit **480**(B) adjusts the inductance of the one or more variable inductance components **485**(B) to adjust/change the resonant (tuned) frequency of the implantable resonant circuit **442**(B). It is to be appreciated that these two techniques for adjusting the resonant frequency of the internal resonant are merely illustrative and that the resonant frequency of an implantable resonant circuit may be changed in other manners.

(68) FIGS. **4**A and **4**B illustrate embodiments in which the tuned frequencies of implantable resonant circuits are adjusted/tuned to a substantially same frequency as a coupled external resonant circuit. FIG. **4**C illustrates an alternative embodiment in which an external resonant circuit can be tuned to match the frequency of an implantable resonant circuit.

(69) More specifically, shown in FIG. **4**C is a portion of an external component **402**(C) and an implantable component **404**(C), in accordance with embodiments presented herein. The external component **402**(C) comprises an external resonant circuit **440**(C), while the implantable component **404**(C) comprises an implantable resonant circuit **442**(C). Collectively, the external resonant circuit **440**(C) and the implantable resonant circuit **442**(C) for a resonant system **450**(C) for use in the transcutaneous transfer of power and data, in accordance with embodiments presented herein.

(70) The external resonant circuit **440**(C) comprises, among other elements, an external coil **406**(C) and capacitive components **487**(C), including one or more variable capacitance components **481**(C). The implantable resonant circuit **442**(C) comprises, among other elements, an internal coil **422**(C) and capacitive components **489**(C). Electrically coupled to, and potentially forming part of, the implantable resonant circuit **442**(C) is internal RF interface circuitry **424**(C), only a portion of which is shown in FIG. **4**C. The resonant system **450**(C) functions as a bidirectional closely-coupled wireless link, generally illustrated by arrow **427**(C).

(71) Electrically coupled to the external resonant circuit **440**(C) is external RF interface circuitry **421**(C), only a portion of which is shown in FIG. **4**C. The external RF interface circuitry **421**(C) comprises, among other elements, data drive circuitry **444**(C), power drive circuitry **446**(C), and a controller **448**(C). Similar to as described above with reference to FIG. **3**, the data drive circuitry **444**(C) and power drive circuitry **446**(C) may be selectively activated/used, for example under the control of controller **448**, for transcutaneous data and power transmissions via external resonant circuit **440**(C).

(72) More specifically, as noted above, in certain examples power and data are transmitted using a type of time division multiple access (TDMA) technique to share the closely-coupled wireless link **427**(C) formed by resonant system **450**(C) (i.e., the closely-coupled wireless link **427**(C) is used to separately transfer power and data from the external component **402**(C) to the implantable component **404**(C), where the transfer of power and data occur during separate time slots using the same external coil **406**(C)). Therefore, during a set of first time periods, the power drive circuitry

446(C) is configured to drive (energize) the external coil **406**(C) with power drive signals **464**(C). The power drive signals **464**(C) comprise an alternating waveform having a steady base frequency of alternation (i.e., a constant burst of square wave at the frequency of resonance of the coil). The frequency of alternation of the power drive signals **464**(C) is sometimes referred to herein as the "power transmission frequency" or the "first frequency." The first frequency of the power drive signals **464**(C) corresponds to a resonant frequency of the resonant system **450**(C). That is, the first frequency may be substantially the same as the resonant frequency of the resonant system **450**(C).

(73) When the coil **406**(C) is driven with the power drive signals **464**(C), current flow is induced in the implantable coil **422**(C), where the current flow corresponds to (i.e., represents) the power drive signals **464**(C). As such, via the inductive link between coils **406**(C) and **422**(C), the power drive signals **464**(C) are received at the internal RF interface circuitry **424**(C). The internal RF interface circuitry **424**(C) is configured to direct the power drive signals **464**(C) to, for example, an implantable rechargeable battery and/or other components.

(74) During a second set of time periods, the data drive circuitry **444**(C) is configured to drive (energize) the external coil **406**(C) with data drive signals **462**(C) in a manner that sends data to the implantable component. The data drive signals **462**(C) comprise the data to be transmitted (e.g., stimulation control data) that is encoded (modulated) onto a carrier signal (i.e., an alternating waveform having a steady base frequency of alternation), where the carrier signal has a second frequency. The frequency of the data carrier signals (i.e., the frequency of the data drive signals **462**) is frequency spaced from the resonant frequency of the resonant system **450**(C), and is sometimes referred to herein as the "data transmission frequency" or the "second frequency."

(75) The data transmission frequency of the data drive signals **462**(C) is frequency spaced a sufficient distance from the resonant frequency of the link **427**(C) to provide the appropriate quality factor for high bandwidth frequency. The data transmission frequency can be higher or lower than the resonant frequency. In certain embodiments, the data transmission frequency may be a multiple or a division of the resonant frequency.

(76) When the coil **406**(C) is driven with the data drive signals **462**(C), current flow is induced in the implantable coil **422**(C), where the current flow corresponds to (i.e., represents) the data drive signals **462**(C). As such, via the inductive link between coils **406**(C) and **422**(C), the data drive signals **462**(C) are received at the internal RF interface circuitry **424**(C). The internal RF interface circuitry **424**(C) is configured to direct the data drive signals **462**(C) to a data output to which any of a number of other components, which have been omitted from FIG. **4**C for ease of illustration, may be connected.

(77) As shown, the data drive circuitry **444**(C) and the power drive circuitry **446**(C) are connected to the external resonant circuit **440**(C) via a driver circuit **468**(C), of which a number of different arrangements is possible. In the example of FIG. **4**C, the driver circuit **468**(C) comprises a switch **472**(C) and an amplifier **474**(C). However, it is to be appreciated that the arrangement for driver circuit **468**(C) shown in FIG. **4**C is merely illustrative and that a driver circuit in accordance with embodiments presented herein may have any of a number of different arrangements.

(78) In FIG. **4**C, the data drive signals **462**(C) and the power drive signals **464**(C) comprise two inputs to the driver circuit **468**(C). The switch **472**(C) operates under the control of controller **448**(C) (i.e., control signal **470**(C)) to selectively enable the data drive signals **462**(C) or the power drive signals **464**(C) to pass to the amplifier **474**(C).

(79) Resonant systems in accordance with embodiments presented herein that provide a closely-coupled wireless link, such as link **427**(C), are designed to be tuned to maximum power coupling. That is, in accordance with embodiments presented herein, during operation, each of the external resonant circuit **440**(C) and the implantable resonant circuit **442**(C) are configured to be tuned to substantially the same first frequency, where the first frequency provides a high quality factor. Power signals are then transmitted over the closely-coupled wireless link **427**(C) at this same first frequency.

(80) Whereas in FIG. **3** the external resonant circuit and the implantable resonant circuit are pre-tuned to the substantially same first frequency, in the arrangement of FIG. **4**C the external component **402**(C) is configured to dynamically tune the external resonant circuit **440**(C) to a frequency that matches a tuned frequency of the implantable resonant circuit **442**(C). The data drive circuitry **444**(C) and the power drive circuitry **446**(C) can also be programmed based on the tuned frequency of the implantable resonant circuit **442**(C). That is, in the example of FIG. **4**C, the implantable resonant circuit **442**(C) has a pre-tuned frequency. Once the external resonant circuit **440**(C) is coupled to the implantable resonant circuit **442**(C) (i.e., so as to form the resonant system **450**(C)), the external component **402**(C) can determine an appropriate tuning for the external resonant circuit **440**(C), as well as appropriate frequencies for the data drive signals **462**(C) and/or the power drive signals **464**(C).

(81) In the example of FIG. **4**C, the implantable component **404**(C) includes a control circuit **480**(C) that is able to provide feedback **486**(C) to the external component **402**(C). Using the feedback **486**(C), the controller **448**(C) of the external component **402**(C) can adjust the frequency of resonance (i.e., the tuned frequency) of the external resonant circuit **440**(C), as well as appropriate frequencies for the data drive signals **462**(C) and/or the power drive signals **464**(C). For example, in the arrangement of FIG. **4**C, the external resonant circuit **440**(C) includes, among other elements, one or more variable capacitance components **481**(C) collectively having a capacitance that is controlled/set by the controller **448**(C). The capacitance of the one or more variable capacitance components **481**(C) can be adjusted, for example, in an analog manner, with digital chips designed to switch different capacitors into the circuit, or in another manner. By adjusting the capacitance of the one or more variable capacitance components **481**(C) the control circuit **480**(C) can adjust the resonant (tuned) frequency of the external resonant circuit **440**(C) either up or down.

(82) The control circuit **480**(C) is configured to use the power and/or data signals **482**(C) received at the implantable resonant circuit **442**(C) to determine the point of maximum power coupling, and accordingly to generate the feedback **486**(C). The controller **448**(C) can then set (i.e., fix) the external resonant circuit **440**(C) to the correct tuned frequency (i.e., fix the capacitance of the one or more variable capacitance components **481**(C) to a level that achieves the selected tuned frequency for external resonant circuit **440**(C) when the feedback **486**(C) indicates a maximum power coupling. These examples may be advantageous in that, during manufacturing of the system, there will be no “set tuning” stage for at least the implantable resonant circuit **442**(C).

(83) As noted above, in the example of FIG. **4**C, the control circuit **480**(C) is configured determine the tuned frequency of the implantable resonant circuit **442**(C) that provides a maximum power coupling with the external resonant circuit (i.e., when the implantable resonant circuit **442**(C) is tuned to a frequency that substantially matches the tuned frequency of the external resonant circuit). In certain examples, the control circuit **480**(C) includes or is coupled to a measurement circuit **483**(C) that can be used to measure the voltage of the received signals **482**(C) or to use the received signals **482**(C) to determine the power being drawn over the RF link (e.g., switching a resistor and measuring the rectified voltage enables determination of the amount of power drawn). In certain embodiments, the control circuit **480**(C) may generate the feedback **486**(C) so as to cause the controller **448**(C) in the external component **402**(C) to adjust the capacitance of the one or more variable capacitance components **481**(C) in a manner that increases or decreases the tuned frequency of the external resonant circuit **440**(C) in a manner that increases the power measured at the measurement circuit **483**(C). The control circuit **480**(C) continues to generate feedback **486**(C) causing such adjustments until a decrease in the power measured at the measurement circuit **483**(C) is detected. At this point, the control circuit **480**(C) generates feedback **486**(C) causing a reversal in the adjustment to again increase the power. Using increasingly smaller up and down adjustments to the tuned frequency (i.e., to the capacitance of the one or more variable capacitance components **481**(C)), the controller **448**(C) can, using feedback from the control circuit **480**(C), accurately lock

to the correct tuned frequency (e.g., to a substantially same frequency as that of the implantable resonant circuit). Once this dynamic tuning is completed, the external resonant circuit **440**(C) remains tuned to the tuned frequency (e.g., the same first) during receipt of both power and data from the external component.

(84) FIG. **4**C illustrates an example in which the tuned frequency for the external resonant circuit **440**(C) is determined based on the point of maximum power coupling. Similar to the above embodiments, the tuned frequency for the external resonant circuit **440**(C) could alternatively be set based on a measurement the frequency of the received signals **482**(C) or in another manner. In a still other alternative arrangement, the control circuit **480**(C) and controller **448**(C) could operate to adjust the tuned frequency of the external resonant circuit **440**(C), not to maximum power coupling, but to a non-optimized power coupling.

(85) As noted, in the example of FIG. **4**C, the controller **448**(C)) adjusts the capacitance of the one or more variable capacitance components **481**(C) to adjust/change the resonant (tuned) frequency of the implantable resonant circuit **442**(C). It is to be appreciated that this technique for adjusting the resonant frequency of the internal resonant is merely illustrative and that the resonant frequency of an implantable resonant circuit may be changed in other manners (e.g., adjustable/variable inductance, etc.).

(86) FIG. **5** is a flowchart of a method **590** in accordance with embodiments presented herein. Method **590** begins at **592** where, during a first set of time periods, circuitry drives an external resonant circuit comprising an external coil with power drive signals having a first center frequency to cause the external coil to transfer power to an implantable resonant circuit. At **594**, during a second set of time periods that are different from the first set of time periods, the circuitry drives the external resonant circuit with data drive signals having a second center frequency to cause the external coil to transfer data to the implantable resonant circuit. The second frequency is different from the first frequency, and the external resonant circuit and the implantable resonant circuit each have an associated tuned frequency that remains the same during each of the first and second sets of time periods.

(87) FIG. **6** is a flowchart of a method **690** in accordance with embodiments presented herein. Method **690** begins at **592** where an external resonant circuit of an external component of an implantable medical devices sends power signals to an implantable resonant circuit of the implantable medical device, wherein the power signals have a first frequency. At **694**, the external resonant circuit sends data signals to the implantable resonant circuit, wherein the data signals have a second frequency. A physical arrangement of each of the implantable resonant circuit and the external resonant circuit does not change whether sending the power or data signals to the implantable resonant circuit.

(88) As noted above, merely for purposes of illustration, the techniques presented herein have been described with reference to a cochlear implant having an external component and an implantable component. However, it is to be appreciated that the techniques presented herein any be implemented in any of a number of different types of implantable medical device systems in which power and data are transferred over a transcutaneous communication link. For example, the techniques presented herein may be used in any other partially or fully implantable medical devices now known or later developed, including other auditory prostheses, such as auditory brainstem stimulators, electro-acoustic hearing prostheses, acoustic hearing aids, bone conduction devices, middle ear prostheses, direct cochlear stimulators, bimodal hearing prostheses, etc. The techniques presented herein may also be used with balance prostheses (e.g., vestibular implants), retinal or other visual prosthesis/stimulators, occipital cortex implants, sensor systems, implantable pacemakers, drug delivery systems, defibrillators, catheters, seizure devices (e.g., devices for monitoring and/or treating epileptic events), sleep apnea devices, electroporation devices, spinal cord stimulators, deep brain stimulators, motor cortex stimulators, sacral nerve stimulators, pudendal nerve stimulators, vagus/vagal nerve stimulators, trigeminal nerve stimulators, diaphragm

(phrenic) pacers, pain relief stimulators, other neural, neuromuscular, or functional stimulators, etc.
(89) FIG. **7** is a schematic diagram illustrating a balance prosthesis in which the techniques presented herein may be implemented. It is to be appreciated is merely illustrative one additional type of implantable medical device in which the techniques presented herein may be implemented.
(90) More specifically, certain individuals may suffer from a balance disorder with complete or partial loss of vestibular system function/sensation in one or both ears. In general, a balance disorder is a condition in which an individual lacks the ability to control and/or maintain a proper (balanced) body position in a comfortable manner (i.e., the recipient experiences some sensation(s) of disbalance). Disbalance, sometimes referred to herein as balance problems, can manifest in a number of different manners, such as feelings of unsteadiness or dizziness, a feeling of movement, spinning, or floating, even though standing still or lying down, falling, difficulty walking in darkness without falling, blurred or unsteady vision, inability to stand or walk un-aided, etc. Balance disorders can be caused by certain health conditions, medications, aging, infections, head injuries, problems in the inner ear, problems with brain or the heart, problems with blood circulation, etc. In general, a “balance prosthesis” or “balance implant” is a medical device that is configured to assist recipients (i.e., persons in which a balance prosthesis is implanted) that suffer from balance disorders.
(91) As noted, FIG. **7** illustrates one example balance prosthesis, namely a vestibular nerve stimulator **700**, in accordance with embodiments presented herein. More specifically, as shown in FIG. **7**, the vestibular nerve stimulator **700** comprises an external component **702** and an implantable component **704**, which is implantable within a recipient (i.e., implanted under the skin/tissue **705** of a recipient).
(92) The external component **702** may comprise a number of functional and/or electronic elements used in the operation of the vestibular nerve stimulator **700**. However, for ease of understanding, FIG. **7** only illustrates external radio frequency (RF) interface circuitry **721** and an external coil **706**. The external coil **706** is part of an external resonant circuit **740**. As described further below, the external RF interface circuitry **721** comprises data drive circuitry **744** and power drive circuitry **746** which are selectively activated/used for transcutaneous transmissions of data and power, respectively, to the implantable component **704**.
(93) The implantable component **704** comprises an implant body (main module) **714** and a vestibular stimulation arrangement **737**. The implant body **734** generally comprises a hermetically-sealed housing **715** in which a number of functional and/or electronic elements used in the operation of the vestibular nerve stimulator **700** may be disposed. However, for ease of understanding, FIG. **7** only illustrates internal radio frequency (RF) interface circuitry **724**, a stimulator unit **720**, and a rechargeable battery **729**. The implant body **734** also includes an internal/implantable coil **722** that is generally external to the housing **715**, but which is connected to the internal RF interface circuitry **724** via a hermetic feedthrough (not shown in FIG. **7**). The implantable coil **722** is part of an implantable resonant circuit **742**. The stimulator unit **720** may include, for example, one or more current sources, switches, etc., that collectively operate to generate and deliver the electrical stimulation signals to the recipient via the vestibular stimulation arrangement **737**.
(94) As shown in FIG. **7**, the vestibular stimulation arrangement **737** comprises a lead **716** and a vestibular nerve stimulating (electrode) assembly **718**. The stimulating assembly **718** comprises a plurality of electrodes **726** disposed in a carrier member **734** (e.g., a flexible silicone body). In this specific example, the stimulating assembly **718** comprises three (3) electrodes, referred to as electrodes **726**(**1**), **726**(**2**), and **726**(**3**). The electrodes **726**(**1**), **726**(**2**), and **726**(**3**) function as an electrical interface to the recipient's vestibular nerve. It is to be appreciated that this specific embodiment with three electrodes is merely illustrative and that the techniques presented herein may be used with stimulating assemblies having different numbers of electrodes, stimulating assemblies having different lengths, etc.

(95) The stimulating assembly **718** is configured such that a surgeon can implant the stimulating assembly adjacent the otolith organs of the peripheral vestibular system via, for example, the recipient's oval window. That is, the stimulating assembly **718** has sufficient stiffness and dynamics such that the stimulating assembly can be inserted through the oval window and placed reliably within the bony labyrinth adjacent the otolith organs (e.g., sufficient stiffness to insert the stimulating assembly to the desired depth between the bony labyrinth and the membranous labyrinth).

(96) As noted above, the external component **702** comprises an external resonant circuit **740**, which includes the external coil **706**. Similarly, the implantable component **704** comprises an implantable resonant circuit **742**, which includes the implantable coil **722**. When the coils **706** and **722** are positioned in close proximity to one another, the coils form a transcutaneous closely-coupled wireless link **727**. This closely-coupled wireless link **727** formed between the external coil **706** with the implantable coil **722** may be used to transfer power and/or data from the external component **702** to the implantable component **704**. In certain examples, the power and data are transmitted using a type of time division multiple access (TDMA) technique to share the closely-coupled wireless link **727**. That is, the closely-coupled wireless link **727** is used to separately transfer power and data from the external component **702** to the implantable component **704**, where the transfer of power and data occur during separate (different and non-overlapping) time slots using the same external coil **706** (i.e., a shared external coil for both data and power). For example, during a set of first time periods, the power drive circuitry **746** of the external RF interface circuitry **721** is configured to drive (energize) the external coil **706** in a manner that sends data to the implantable component **104**. During a second set of time periods, the data drive circuitry **744** of the external RF interface circuitry **721** is configured to drive (energize) the external coil **706** in a manner that sends power to the implantable component **704**.

(97) As noted, in the example of FIG. **7**, the external coil **706** is part of an external resonant circuit (e.g., external resonant tank circuit) **740**. Similarly, the implantable coil **722** and at least a portion of the internal RF interface circuitry **724** form an implantable resonant circuit (e.g., internal resonant tank circuit) **742**. The external resonant tank circuit **740** and the internal resonant tank circuit **742** collectively form a resonant system **750** which function as closely-coupled wireless link **727**.

(98) In order to efficiently transfer power from the external component to **702** to the implantable component **704**, the closely-coupled wireless link **727** (resonant system **750**) should have a high quality factor. That is, the quality factor of the closely-coupled wireless link **727** should be maximized during power transmission, thereby ensuring low power loss. However, as noted, a high quality factor is associated with a narrow bandwidth, which is problematic for transmission of data over the closely-coupled wireless link **727**. Therefore, power transmission and data transmission have competing quality factor requirements (i.e., efficient power transmission requires a high/maximum quality factor, while higher bandwidth data requires a lower quality factor).

(99) The techniques presented herein address these competing quality factor requirements for power and data transmissions through the use of different transmit (drive) frequencies at the external resonant circuit **740**. More specifically, in the embodiments of FIG. **7**, the power drive circuitry **746** is configured to drive the external coil **706** (external resonant inductive coil) at a first frequency to transmit power over the closely-coupled wireless link **727**. Both the external resonant circuit **740** and the implantable resonant circuit **742** are substantially tuned to this same first frequency. That is, the external resonant circuit **740** and the implantable resonant circuit **742** are each structurally configured so as to resonate a frequency that is substantially the same as the first frequency. Accordingly, the resonant system **750** may be referred to as being tuned to the first frequency. In other words, in these embodiments, the first frequency for power transmission is the resonant frequency of the resonant system **750** (i.e., the resonant frequency of each of the external resonant circuit **740** and the implantable resonant circuit **742**).

(100) Due to the fact that the power transmissions occur at a frequency that substantially matches the tuned frequency of each of the external resonant circuit **740** and the implantable resonant circuit **742** (i.e., the tuned frequency of the resonant system **750**), maximum power coupling is achieved with the power transmissions at the first frequency. Stated differently, the matching of the drive/transmit frequency to the tuned frequency of the resonant system **750** provides a high quality factor where, as noted, the higher the quality factor of the system, the more efficient the power transfer will be across the closely-coupled wireless link **727**.
(101) While, as noted above, a high quality factor is appropriate for power transmission, a high quality factor reduces the rate that data can be transmitted through the inductive coupling of the closely-coupled wireless link **727** (i.e., reduces the available bandwidth of the closely-coupled wireless link **727**). While the quality factor of the resonant system is high when the transmit frequency is at or near the resonant frequency, the quality factor is lower at different frequencies that have an appropriate distance/spacing, in frequency, from the resonant frequency. Accordingly, an appropriate quality factor for transmitting data can be obtained at transmit frequencies that are spaced some frequency distance from the resonant frequency of the resonant system **750**. Therefore, in accordance with embodiments presented herein, data drive circuitry **744** is configured to drive the external resonant circuit **740**, including external coil **706**, at a second frequency to transmit data over the closely-coupled wireless link **727**, where the second frequency is different from the first frequency. During the data transmission, the external resonant circuit **740** and the implantable resonant circuit **742** both remain tuned to the first frequency (i.e., the external resonant circuit **740** and the implantable resonant circuit **742** each have a fixed structure that fixes the tuned frequency thereof). As such, the frequency "mismatch" or difference between the transmit frequency and the frequency of the resonant system **750** causes a reduction in the quality factor of the combined resonant system (i.e., reduces the quality factor of the closely-coupled wireless link **727**), which in turn increases the bandwidth available for the transmission of the data.
(102) In summary, FIG. **7** illustrates an arrangement in accordance with embodiments presented herein in which, during a first set of time periods, the external resonant circuit **740**, which includes external coil **706**, is driven at a first frequency to transmit power to the implantable resonant circuit **742**, including implantable coil **722**. During a second set of time periods, the external resonant circuit **740** is driven at a second frequency to transmit data to the implantable resonant circuit **742**, where the second frequency is frequency spaced a frequency distance from the first frequency. During both the first and second sets of time periods, the external resonant circuit **740** and the implantable resonant circuit **742** remain tuned to the first frequency (i.e., the external resonant circuit **740** and the implantable resonant circuit **742** have a fixed tuning).
(103) It is to be appreciated that the embodiments presented herein are not mutually exclusive.
(104) The invention described and claimed herein is not to be limited in scope by the specific preferred embodiments herein disclosed, since these embodiments are intended as illustrations, and not limitations, of several aspects of the invention. Any equivalent embodiments are intended to be within the scope of this invention. Indeed, various modifications of the invention in addition to those shown and described herein will become apparent to those skilled in the art from the foregoing description. Such modifications are also intended to fall within the scope of the appended claims.

Claims

1. An implantable medical device, comprising: an implantable resonant circuit comprising an implantable coil; an external resonant circuit comprising an external coil configured to transcutaneously transfer power and data to the implantable resonant circuit using separate power and data time slots, respectively; and external radio-frequency (RF) interface circuitry configured to drive the external resonant circuit at a first frequency during the power time slots and to drive the

external resonant circuit at a second frequency during the data time slots, wherein the second frequency is different from the first frequency.

2. The implantable medical device of claim 1, wherein the implantable resonant circuit and the external resonant circuit each have a fixed resonant frequency that does not change during either of the power or the data time slots.

3. The implantable medical device of claim 2, wherein the fixed resonant frequencies of the implantable resonant circuit and the external resonant circuit are substantially the same as the first frequency.

4. The implantable medical device of claim 1, wherein a physical arrangement of each of the implantable resonant circuit and the external resonant circuit remains fixed during each of the power and the data time slots.

5. The implantable medical device of claim 1, wherein the first frequency is selected based on a resonant frequency of each of the implantable resonant circuit and the external resonant circuit and is a frequency that corresponds to a predetermined power coupling between the external resonant circuit and the implantable resonant circuit.

6. The implantable medical device of claim 5, wherein the predetermined power coupling is a substantially maximum power coupling between the external resonant circuit and the implantable resonant circuit.

7. The implantable medical device of claim 5, wherein the predetermined power coupling is a non-optimized power coupling between the external resonant circuit and the implantable resonant circuit.

8. The implantable medical device of claim 1, wherein the first frequency provides a selected power coupling between the external resonant circuit and the implantable resonant circuit, and wherein the second frequency is frequency spaced from the first frequency so as to provide a selected bandwidth for transfer of data to the implantable coil during the data time slots.

9. The implantable medical device of claim 1, wherein the second frequency is spaced from the first frequency by a predetermined frequency distance.

10. The implantable medical device of claim 1, further comprising: a control circuit coupled to the implantable resonant circuit, wherein the control circuit is configured to: identify a maximum power coupling between the external resonant circuit and the implantable resonant circuit; and set a resonant frequency of the implantable resonant circuit based on the identified maximum power coupling.

11. The implantable medical device of claim 10, wherein the implantable resonant circuit comprises one or more variable capacitance components, and wherein the control circuit is configured to adjust the capacitance of the one or more variable capacitance components to set the resonant frequency of the implantable resonant circuit based on the resonant frequency of the external resonant circuit.

12. The implantable medical device of claim 10, wherein the control circuit is configured to set the resonant frequency of the implantable resonant circuit to a frequency that is substantially the same as the resonant frequency of the external resonant circuit.

13. The implantable medical device of claim 1, wherein the first frequency is selected based on a resonant frequency of each of the implantable resonant circuit and the external resonant circuit and is a frequency that corresponds to a non-optimized power coupling between the external resonant circuit and the implantable resonant circuit.

14. The implantable medical device of claim 1, wherein the second frequency is a multiple of the first frequency.

15. The implantable medical device of claim 1, wherein the second frequency is a division of the first frequency.